

algorithms to solve them. We'll conclude by discussing some of the deepest mysteries of the world of algorithms, including detailed instructions on how to win a million dollars with advanced algorithmic theory.

Doing More with Algorithms

The 10 previous chapters of this book covered algorithms that can perform a variety of tasks in many fields. But algorithms can do even more than we've seen here. If you wish to continue your adventure with algorithms, you should explore other fields and the important algorithms associated with them.

For example, the many algorithms for information compression can store a long book in a coded form that is only a fraction of the size of the original, and they can compress a complex photograph or film file into a manageable size with either minimal or no loss of quality.

Our ability to communicate securely online, including confidently passing our credit card information to third parties, relies on cryptographic algorithms. Cryptography is great fun to study because it comes with a thrilling history of adventurers, spies, betrayals, and triumphant nerds who broke codes to win wars.

Recently, innovative algorithms have been developed to perform parallel distributed computing. Instead of performing one operation at a time, millions of times, distributed computing algorithms split up a dataset into many little parts and then send them to different computers, which perform the needed operation simultaneously and return the results, to be recompiled and presented as the final output. By working on all parts of the data concurrently instead of consecutively, parallel computing saves a huge amount of time. This is extremely useful for applications in machine learning, where there's a need to process datasets that are extremely large or to perform a large number of simple computations simultaneously.

For decades, people have been excited about the potential of quantum computing. Quantum computers, if we can engineer them to work properly, have the potential to perform extremely difficult calculations (including the calculations needed to break state-of-the-art cryptography) in a tiny fraction of the time required on today's nonquantum supercomputers. Since